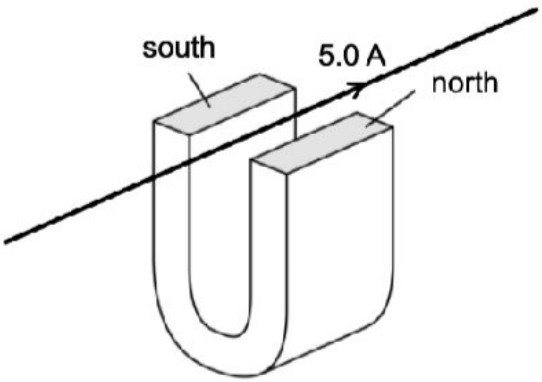
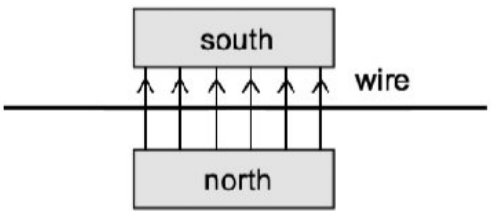
25 A wire connected to a d.c. power supply is positioned between the poles of a U-shaped magnet. You may assume that the wire is at right angles to a uniform magnetic field between the poles and there is no magnetic field outside the poles.



(not to scale)

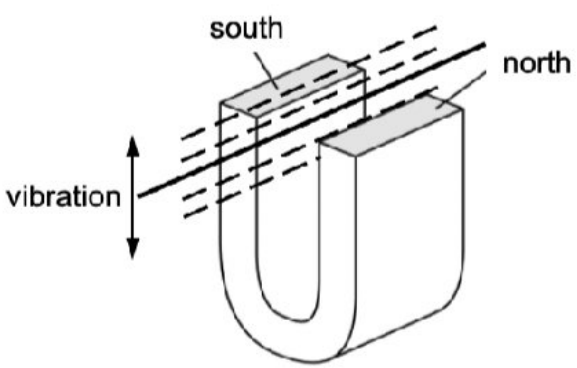


(top view)

When the current in the wire is 5.0 A, the force acting on the wire due to the magnetic field is 10 mN.

The wire is now disconnected from the power supply.

The wire is made to vibrate vertically with an amplitude of 1.8×10^{-3} m and a frequency of 90 Hz. You may assume that the segment of the wire between the poles to remain horizontal during the vibration as shown below.



What is the maximum e.m.f. induced across the wire?

A 0.32 mV **B** 2.0 mV **C** 0.32 V **D** 2.0 V